

Machine Learning Analysis of Organic Co-Solvents for Water-in-Salt Zinc-Ion Battery Electrolytes: A Cross-Study Dataset and Predictive Framework

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Abstract—Electrolyte design for aqueous Zn batteries is still largely empirical. We assembled a cross-study dataset of 108 water-in-salt Zn electrolyte formulations from 63 open-access papers published between 2018 and early 2026, and benchmarked four regressors using 23 molecular, physicochemical, and formulation features. Random Forest gave the best cycle-lifespan model with cross-validated $R^2 = 0.51$ ($p = 0.01$), while XGBoost performed best for ionic conductivity with $R^2 = 0.30$ ($p = 0.01$). SHAP ranks co-solvent fraction and the DN $\times$ fraction term above donor number alone, consistent with an effective donor concentration picture. CE and ESW remained non-predictive ($R^2 < 0$), which is consistent with uneven reporting across the source literature. Restricting optimization to lifespan and conductivity highlights DMSO, DMF, and triethyl phosphate for follow-up. This conference version focuses on the validated cross-study trends and the resulting solvent ranking.

Index Terms—zinc-ion battery, machine learning, water-in-salt electrolyte, SHAP analysis, co-solvent

I. Introduction

Grid-scale energy storage needs to be cheaper, safer, and more scalable than current lithium-ion technology. Lithium-ion batteries dominate the market, but lithium scarcity, safety risks, and environmental concerns drive the search for alternatives [1], [2]. Aqueous zinc-ion batteries (AZIBs) offer high theoretical capacity (820 mAh/g for Zn metal anodes), intrinsic safety from non-flammable aqueous electrolytes, low cost, and environmental benignity [3]–[5].

AZIBs face electrochemical challenges that hinder broader deployment. The narrow electrochemical stability window of water (~ 1.23 V) limits achievable energy density, while parasitic hydrogen evolution reaction (HER) and zinc dendrite formation compromise cycling stability [6], [7]. Electrolyte engineering can address these problems. Water-in-salt (WIS) electrolytes, first demonstrated by Suo et al. [8] for lithium systems, suppress water activity through high salt concentrations, expanding the ESW to > 3 V in some formulations [5], [9].

Hybrid electrolytes with organic co-solvents improve on the WIS concept. We previously showed that acetonitrile-water-in-salt (AWIS) electrolytes achieve 3.4 V stability windows while maintaining sufficient ionic conductivity [10]. Since then, researchers have explored co-solvents including dimethyl sulfoxide (DMSO), ethylene glycol,

glycerol, and phosphate esters, each offering distinct advantages in suppressing HER, modifying solvation structures, or enhancing Zn plating/stripping reversibility [11]–[13].

Co-solvent selection remains empirical. Luo et al. [14] showed that donor number correlates with additive effectiveness in dilute aqueous electrolytes, reaching $R^2 \approx 0.6$ for Zn plating efficiency, but their study focused on a single metric and did not address multi-objective prediction in WIS formulations.

Related ML studies have explored self-supervised, sequence, and graph/boosting models [15]–[17], but cross-study WIS zinc datasets remain scarce. Here we assemble a 108-formulation dataset spanning more than 20 co-solvents and salt concentrations from 0.5 to 30 molal, test four ML models for lifespan, CE, ESW, and ionic conductivity, and use SHAP and Pareto screening to interpret the design space.

II. Methodology

A. Dataset Construction

Google Scholar and OpenAlex searches targeted publications from 2018 through early 2026 using search terms such as “zinc-ion battery,” “aqueous electrolyte,” “co-solvent,” and “water-in-salt.” The dataset was restricted to open-access publications with clearly reported experimental conditions.

Extracted data included co-solvent identity (SMILES or systematic name), salt type and concentration, co-solvent fraction, and performance metrics (lifespan, CE, ESW, conductivity). When multiple metrics were reported (e.g., CE at different cycle numbers), values at 100 cycles or the closest available cycle number were used.

The raw extraction contained 174 entries from 63 source publications; 31 duplicate records were removed using a DOI/co-solvent/salt-concentration key. RDKit [18] accepted 82 reported SMILES directly, and 12 more were recovered by manual PubChem lookup. Most of the remaining 49 unresolved cases were records without a discrete co-solvent identity, names that could not be mapped to validated canonical SMILES, or multi-component formulations with at least one unresolved

component. The final dataset comprised 108 electrolyte formulations with varying completeness across features.

B. Feature Engineering

We used 23 features in three groups.

Molecular Descriptors (11 features). From validated SMILES using RDKit (v2023.09): molecular weight (MolWt), octanol-water partition coefficient (LogP), topological polar surface area (TPSA), hydrogen bond donor/acceptor counts, rotatable bond count, aromatic ring count, fraction of sp^3 carbons, molar refractivity (MolMR), heteroatom count, and total ring count.

Physicochemical Properties (6 features). Literature values for donor number (DN, kcal/mol), acceptor number (AN), dielectric constant (ϵ), viscosity (η , cP at 25°C), boiling point, and dipole moment. DN and AN quantify Lewis basicity and acidity, which govern ion solvation [19]. Missing values were imputed using median substitution; approximately 40% of physicochemical features required imputation.

Formulation Features (6 features). Salt concentration (molal) and co-solvent volume fraction were extracted from source publications. Four interaction terms were engineered: DN $\times$ cosolvent_fraction, dielectric $\times$ salt_concentration, viscosity $\times$ salt_concentration, and LogP $\times$ dielectric. Dataset coverage was 67% for molecular descriptors, 62–85% for physicochemical properties, 61–85% for formulation variables, and 49% for interaction terms.

Lifespan and conductivity exhibited right-skewed distributions spanning multiple orders of magnitude; $\log_{10}$ transformation was applied. All features were standardized using z-score normalization.

C. Machine Learning Models

Four regression algorithms were compared: Gaussian Process Regression (GPR) with a Matérn 5/2 kernel as a probabilistic baseline, XGBoost [20] with 200 estimators, depth 4, and learning rate 0.1, Random Forest [21] with 300 estimators and default scikit-learn settings, and Ridge Regression with L2 regularization ($\alpha = 1.0$).

D. Model Validation

Model generalization was assessed in three ways. Five-fold cross-validation reports the mean and standard deviation of R^2 across held-out folds. Leave-one-paper-out (LOPO) cross-validation treats each source publication as a fold, simulating prediction for an unseen laboratory and preventing records from the same paper from appearing in both training and test sets. Statistical significance was evaluated with 200 permutation tests by shuffling target labels and comparing the resulting R^2 values with the true score. This makes LOPO the stricter test of whether the models transfer beyond paper-specific reporting patterns.

E. Interpretability Analysis

SHAP (SHapley Additive exPlanations) [22] provided model interpretability. For tree-based models, TreeExplainer was used; for GPR, KernelExplainer with 100 background samples. SHAP values quantify each feature’s contribution to individual predictions.

F. Multi-Objective Pareto Optimization

NSGA-II [23] generated Pareto-optimal co-solvent candidates. Because CE and ESW models have negative CV R^2 and fail permutation tests ($p > 0.05$), Pareto optimization is restricted to lifespan and conductivity:

$$\begin{aligned} \max f_1(\mathbf{x}) &= \hat{y}_{\text{lifespan}}(\mathbf{x}) \\ \max f_2(\mathbf{x}) &= \hat{y}_{\text{conductivity}}(\mathbf{x}) \\ \text{s.t. BP} &> 60^\circ\text{C} \end{aligned}$$

NSGA-II was run for 100 generations with population size 200, yielding approximately 100 Pareto-optimal solutions. Once CE and ESW become predictive, the same framework can be extended to four-objective optimization or threshold-constrained ranking [24].

III. Results and Discussion

A. Dataset Characteristics

The most frequently studied co-solvents are ethylene glycol (7 entries), DMSO (6), glycerol (5), and triethyl phosphate (5). ZnSO_4 -based electrolytes predominate (48 entries), followed by $\text{Zn}(\text{OTf})_2$ (31) and ZnCl_2 (18). Lifespan ranges from 2 to over 5000 hours (median: 200 h), and ionic conductivity spans 0.1–120 mS/cm.

B. Model Performance

Predictive performance differed sharply by target (Table I).

TABLE I
Best-model summary for each target. All CE/ESW models had negative CV R^2 .

Target	Best model	CV R^2	LOPO R^2	p
Lifespan	RF	0.51 ± 0.20	0.477	0.01
Conductivity	XGB	0.30 ± 0.40	0.453	0.01
CE	GPR	-0.38 ± 0.67	0.021	0.33
ESW	RF	-0.29 ± 0.42	-0.456	0.13

Lifespan prediction. Random Forest achieves the best cross-validated performance with $R^2 = 0.507 \pm 0.20$, substantially exceeding the Ridge regression baseline ($R^2 = 0.20$) and demonstrating statistical significance ($p = 0.01$). The gap between training R^2 (0.823) and CV R^2 (0.507) indicates moderate overfitting. The LOPO R^2 of 0.477 suggests that the lifespan model retains predictive value across laboratories.

Conductivity prediction. XGBoost reaches CV $R^2 = 0.298 \pm 0.40$ and LOPO $R^2 = 0.453$ ($p = 0.01$). The stronger LOPO result is consistent with conductivity

being measured more consistently across papers than cycling-based metrics.

Context against prior work. Luo and coworkers [14] reported $R^2 \approx 0.6$ for donor number versus Zn plating efficiency in single-laboratory, dilute-regime data. Our lower but still significant cross-study results are consistent with the additional variability introduced by combining measurements across different laboratories, salts, and co-solvent chemistries.

CE and ESW prediction failure. Both targets exhibit negative cross-validated R^2 values across all models, indicating predictions worse than the mean. We attribute this to tightly clustered CE values (98–99.5%), inconsistent CE definitions (initial-cycle vs. 100th-cycle vs. average), and variable ESW criteria (current density threshold from 0.01 to 1 mA/cm²). Future studies should report cell format, areal loading, cycle number used for CE, current density, cutoff voltage, ESW scan rate, and threshold criterion in a standardized way.

Learning-curve takeaway. Additional learning-curve analysis indicates lifespan performance improving toward $R^2 \approx 0.43$ by ~ 82 samples, whereas conductivity remains unstable near zero at current sample size, indicating that standardized data growth matters as much as model choice.

C. Feature Importance and Design Rules

SHAP ranks the predictors differently for lifespan and conductivity, so we focus on those two targets (Figure 1).

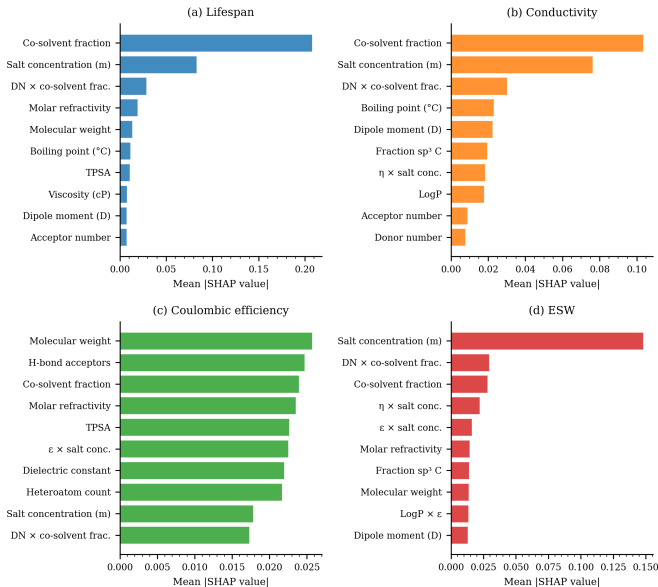


Fig. 1. SHAP summary and cross-target comparison for lifespan and conductivity.

Universal predictors. Two features appear among the top-5 importance rankings for both targets:

Cosolvent fraction is the most important feature for both lifespan and conductivity. Higher co-solvent fractions generally correlate with improved lifespan, consistent with

reduced water activity and HER suppression. For conductivity, the relationship is non-monotonic, with moderate fractions (20–50 vol%) optimal.

Salt concentration ranks second for both targets. In the WIS regime (>15 m), increased concentration improves stability but decreases conductivity due to elevated viscosity.

Donor number interactions. Donor number alone does not rank among the top-5 predictors for lifespan or conductivity (it ranks 10th for conductivity). However, $\text{DN} \times \text{cosolvent_fraction}$ ranks 3rd for both targets. This interaction feature captures what we term “effective donor concentration”—the abundance of high-DN molecules available to coordinate Zn^{2+} .

For example, a co-solvent with $\text{DN} = 30$ at 10 vol% contributes less to Zn^{2+} solvation than one with $\text{DN} = 20$ at 50 vol%. In this dataset, the product term captures that effect better than donor number alone.

Target-specific predictors. Boiling point ranks higher for conductivity than for lifespan, while molar refractivity is more important for lifespan. Together, these results suggest that WIS electrolyte design should favor moderate-to-high-DN co-solvents at 30–50 vol%, while still accounting for conductivity penalties at higher salt concentration.

D. Pareto-Optimal Co-Solvent Profiles

NSGA-II optimization on the two-objective problem generated 100 Pareto-optimal solutions (Figure 2). Pareto-optimal co-solvents cluster around: high donor number (>20 kcal/mol), moderate molecular weight (80–160 g/mol), low-to-moderate LogP (-0.5 to 0.5), and moderate viscosity (0.5 – 3 cP).

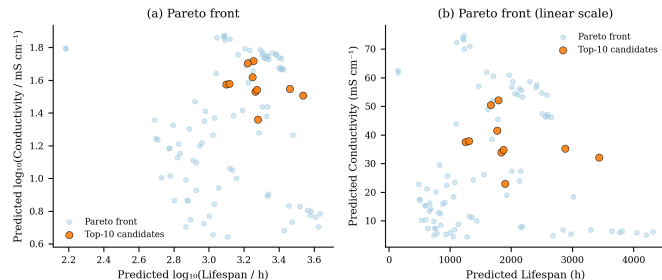


Fig. 2. Pareto front for predicted lifespan and conductivity.

Candidate molecules. Mapping the Pareto region onto known solvents highlights DMSO ($\text{DN} = 29.8$, $\text{LogP} = -1.35$) as an internal check, DMF ($\text{DN} = 26.6$, $\text{LogP} = -1.01$) as a lower-viscosity analogue with limited WIS zinc study, and triethyl phosphate (TEP, $\text{DN} = 24.5$, $\text{LogP} = 0.39$) as a higher-safety candidate supported by five favorable entries. NMP ($\text{DN} = 27.3$, $\text{LogP} = -0.46$) appears only through extrapolation and still requires direct testing. A practical next experiment would compare DMSO, DMF, and TEP at matched salt concentration and co-solvent fraction, which would directly test whether the ranking found here holds under a common protocol.

Relation to prior AWIS work. Acetonitrile, used in our earlier AWIS study [10], does not sit on the Pareto front, but its position remains informative: relatively low donor number can be partly offset by low molecular weight and high accessible fraction. That comparison helps explain why the present screen favors stronger donors such as DMSO, DMF, and TEP in the broader WIS design space.

IV. Conclusion

In this mixed-literature dataset, only lifespan and conductivity were predictable at useful levels. Random Forest performed best for lifespan, and XGBoost performed best for conductivity. CE and ESW did not generalize across studies, likely because the underlying measurements were reported too inconsistently from paper to paper.

SHAP places solvent fraction and $\text{DN} \times \text{cosolvent fraction}$ ahead of donor number alone, and the Pareto screen therefore favors solvents such as DMSO, DMF, and TEP. In practice, the present models are best used as a ranking tool for candidate co-solvents rather than as precise predictors of absolute device performance. DMF and NMP should be treated as out-of-dataset candidates that still need direct experimental validation. Main limitations are dataset size, median imputation for about 40% of physicochemical properties, and missing protocol-level variables; the dataset is also much smaller than corpora typically used for representation learning or multimodal physics-constrained prediction [15], [17], [25]. Taken together, the results provide a compact cross-study screening framework for prioritizing co-solvents in WIS zinc electrolytes.

References

- [1] B. Dunn, H. Kamath, and J.-M. Tarascon, "Electrical energy storage for the grid: A battery of choices," *Science*, vol. 334, no. 6058, pp. 928–935, 2011.
- [2] D. Larcher and J.-M. Tarascon, "Towards greener and more sustainable batteries for electrical energy storage," *Nat. Chem.*, vol. 7, no. 1, pp. 19–29, 2015.
- [3] G. Fang, J. Zhou, A. Pan, and S. Liang, "Recent advances in aqueous zinc-ion batteries," *ACS Energy Lett.*, vol. 3, no. 10, pp. 2480–2501, 2018.
- [4] X. Jia, C. Liu, Z. G. Neale, J. Yang, and G. Cao, "Active materials for aqueous zinc ion batteries: Synthesis, crystal structure, morphology, and electrochemistry," *Chem. Rev.*, vol. 120, no. 15, pp. 7795–7866, 2020.
- [5] F. Wang et al., "Highly reversible zinc metal anode for aqueous batteries," *Nat. Mater.*, vol. 17, no. 6, pp. 543–549, 2018.
- [6] Q. Zhang, J. Luan, Y. Tang, X. Ji, and H. Wang, "Interfacial design of dendrite-free zinc anodes for aqueous zinc-ion batteries," *Angew. Chem., Int. Ed.*, vol. 59, no. 32, pp. 13180–13191, 2020.
- [7] W. Yang et al., "Hydrated eutectic electrolytes with ligand-oriented solvation shells for long-cycling zinc-organic batteries," *Joule*, vol. 4, no. 7, pp. 1557–1574, 2020.
- [8] L. Suo et al., "'Water-in-salt' electrolyte enables high-voltage aqueous lithium-ion chemistries," *Science*, vol. 350, no. 6263, pp. 938–943, 2015.
- [9] J. Xie, Z. Liang, and Y.-C. Lu, "Molecular crowding electrolytes for high-voltage aqueous batteries," *Nat. Mater.*, vol. 19, no. 9, pp. 1006–1011, 2020.
- [10] X. Song, L. Bai, L. Wang, D. Zeng, and J. Liu, "Enhanced reversibility and electrochemical window of Zn-ion batteries with an acetonitrile/water-in-salt electrolyte," *Chem. Commun.*, vol. 57, no. 10, pp. 1246–1249, 2021.
- [11] L. Cao et al., "Fluorinated interphase enables reversible aqueous zinc battery chemistries," *Nat. Nanotechnol.*, vol. 16, no. 8, pp. 902–910, 2021.
- [12] W. Sun et al., "A rechargeable zinc-air battery based on zinc peroxide chemistry," *Science*, vol. 371, no. 6524, pp. 46–51, 2021.
- [13] S. Liu et al., "Tuning the electrolyte solvation structure to suppress cathode dissolution, water reactivity, and Zn dendrite growth in zinc-ion batteries," *Adv. Funct. Mater.*, vol. 31, no. 42, p. 2104281, 2021.
- [14] Y. Luo, X. Chen, H. Zhang, L. Wang, and J. Li, "Machine learning-guided screening of electrolyte additives for dendrite-free zinc anodes," *ACS Nano*, vol. 19, no. 2, pp. 1847–1859, 2025.
- [15] J. Zhang, "Research on Image Representation Learning Method Based on Self-Supervised Learning," *Int. Sci. Tech. Econ. Res.*, 2026. DOI: 10.71451/ISTAER2616.
- [16] Y. Qin, "Research on Long Sequence Learning Behavior Modeling Based on Transformer-XL," *Int. Sci. Tech. Econ. Res.*, 2026. DOI: 10.71451/ISTAER2613.
- [17] X. Zeng, "Cross-Border Trade Fraud Detection via Integrated Heterogeneous Graph Neural Network and XGBoost," *Int. Sci. Tech. Econ. Res.*, 2026. DOI: 10.71451/ISTAER2603.
- [18] G. Landrum, "RDKit: Open-source cheminformatics," [Online]. Available: <https://www.rdkit.org>. [Accessed: Apr. 2026].
- [19] C. Reichardt and T. Welton, *Solvents and Solvent Effects in Organic Chemistry*, 4th ed. Weinheim, Germany: Wiley-VCH, 2011.
- [20] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery Data Mining*, New York, NY, USA, 2016, pp. 785–794.
- [21] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [22] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30*, Red Hook, NY, USA: Curran Associates, 2017, pp. 4765–4774.
- [23] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182–197, Apr. 2002.
- [24] C. Feng, "Personalized Recommendation Algorithm of Piano Practice Scheme Based on Multi Objective Optimization," *Int. Sci. Tech. Econ. Res.*, 2026. DOI: 10.71451/ISTAER2612.
- [25] Y. Ding and G. Chen, "Joint Prediction Model of Reservoir Parameters Based on Multimodal Transformer Graph Neural Operator Physical Constraint Network," *Int. Sci. Tech. Econ. Res.*, 2026. DOI: 10.71451/ISTAER2604.